Dual-Stage Helical Carbon Capture System

Comprehensive Engineering Summary & Financial Model for Small-to-Medium Industrial Flue Gas Valorization

1. Executive Summary & Project Objective

The Dual-Stage Helical Carbon Capture and Conversion System is an innovative, compact, and low-cost environmental engineering solution engineered specifically for small-to-medium industrial flue gas generators (50 m^3/hr). By combining an intelligent spatial layout with continuous chemical valorization, the system captures hazardous carbon dioxide emissions directly at the source and transforms them into marketable high-grade baking soda ($NaHCO_3$), achieving near-zero environmental discharge and complete financial self-sustainability.

Core Engineering Breakthrough: Spatial Footprint Compression

Traditional carbon capture installations require vast tracts of land to achieve the necessary gas-liquid contact time. This design circumvents land constraints by utilizing a compact 20-meter vertical structural framework housing a dual-helix piping network, effectively folding 1,900 meters of travel path into a highly manageable industrial footprint.

2. Physical Infrastructure & System Architecture

The physical plant is meticulously engineered for structural integrity, corrosion resistance, and fluid-dynamic efficiency under continuous industrial workloads:

Vertical Framework

A 20-meter high structural steel framework designed to support the dynamic weight of liquid-filled piping while withstanding wind loads and localized vibration.

Piping Network

Two distinct 950-meter chemical-grade HDPE or PVC pipelines (total length 1,900 meters) with a uniform 10 cm diameter configured in a smooth dual-helix geometry.

Upstream Filtration Unit

Pre-treatment membrane and particulate separation modules that strip 85% of inert gases (N_2, O_2) prior to reactor entry.

Pumps & Boosters

Two low-power chemical-grade industrial blowers/pumps (totaling ~0.66 kW power draw) managing gas injection and inter-stage fluid recirculation.

3. Fluid Dynamics, Residence Time & Reaction Kinetics

Optimizing mass transfer rates between gaseous CO_2 and aqueous reactants requires precise control over gas velocity and residence duration. The system achieves exceptional performance through a multi-stage flow mechanism:

- **Volume Reduction & Velocity:** Upstream filtration reduces the initial raw flue gas volume (50 m^3/hr) down to pure CO_2 (6.25 m^3/hr). This substantial volumetric drop reduces internal gas velocity to approximately ***0.22 m/s***.
- **Prolonged Residence Time:** Across the 1,900-meter continuous helical path, the gentle flow rate grants an extended gas-liquid contact time of roughly ***143 minutes*** (2.3 hours), facilitating near-complete chemical absorption.
- **Dual-Stage Recirculation:** A secondary booster pump captures trace unreacted gas exiting the top of the first helical pipe and re-injects it into the base of the second pipe, driving overall capture efficiency to nearly ***99%***.

4. Chemical Valorization Pathway

Inside the helical reactor tubes, carbon dioxide reacts continuously with an circulating aqueous sodium carbonate solution, producing high-purity sodium bicarbonate slurry via the following balanced reaction:

$$Na_2CO_3 + CO_2 + H_2O \rightarrow 2NaHCO_3$$

The resulting dense white slurry is periodically drained from the base of the tower and routed to dedicated lined solar evaporation beds. The thermal drying setup processes the slurry into dry, commercial-grade baking soda powder ready for packaging and commercial distribution.

5. Capital Expenditure (CAPEX) Breakdown

The project is exceptionally cost-effective, keeping initial setup expenses well within reach for small and medium industrial facilities without sacrificing structural durability:

EQUIPMENT / COMPONENT DESCRIPTION	SPECIFICATIONS & MATERIAL GRADE	ESTIMATED COST (INR)
Tower Structure & Scaffolding	20m vertical structural steel frame with protective anti-corrosion coating	₹60,000 - ₹1,00,000
Piping Network	1,900m chemical-grade HDPE/PVC pipes (10 cm dia, dual-helix layout)	₹1,80,000 - ₹2,50,000
Gas Filtration & Separation Unit	Upstream membrane filter module for inert gas stripping	₹1,20,000 - ₹2,00,000
Pumps & Blowers	2 chemical-grade low-power blowers/pumps (~0.66 kW total load)	₹35,000 - ₹50,000
Solar Drying Setup	Lined shallow concrete evaporation pans for slurry processing	₹25,000 - ₹40,000
Total Estimated Initial Setup Cost (CAPEX)		**₹4,20,000 - ₹6,40,000**

Operational Sustainability Note

With a minimal monthly running expenditure (~₹45,000 covering power, chemical replenishment, routine maintenance, and operator oversight) easily offset by baking soda commercial sales, this modular system bridges environmental compliance with immediate financial profitability.